

ECON 351 - Diversification & Variance Problems

Screenshots from homework, sample exam, mock midterm, lecture, and practice tests

Sources checked:

- Homework 1 Ch.4: No diversification/variance problems
- Homework 2 Ch.9: No diversification/variance problems
- Sample Questions Ch.4: No diversification/variance problems
- Sample Questions Ch.9: Question 22 - Feng/Nan portfolios (included)
- Mock Midterm 1: Question 29 - John/Marc variance (included)
- Econ351 Lecture Ch.9: Variance, correlation, diversification slides (included)
- Practice Test 1 Q24: Same as Sample Q22
- Practice Test 2 Q24: Same as Mock Q29
- Practice Test 3 Q24: Same as Sample Q22

6 problem sets in this PDF.

Variance, Correlation & Diversification - Formula Sheet

Expected value: $E[x] = p_1 \cdot x_1 + p_2 \cdot x_2 + \dots + p_n \cdot x_n$

Variance: $\text{Var}(x) = \sum (p_i \cdot (x_i - E[x])^2)$

Higher variance = more risk / more volatile outcomes

Correlation: do variables move together? (positive / negative / zero)

Diversification: combine assets that are NOT fully positively correlated

Negatively correlated portfolio -> lower variance than either asset alone

Positively correlated portfolio -> higher variance (less diversification benefit)

Problem 1: D-1

Sample Questions Ch.9 - Question 22 | Practice Tests 1 & 3 Q24
Feng: agriculture (A & B). Nan: airline/oil (C & D).

Question 22:

Feng has all her wealth invested in Company A and in Company B. These are agricultural companies, and their value can go up or down depending on the amount of rainfall this season. There is a 50% chance that it

type of car he purchases and on whether the future price of gas will be high or low. The following table summarizes his profit for each type of car and for the different gas prices.

	Low Prices	High Prices
Regular Car	\$110	\$80
Electric Car	\$120	\$90

Problem 2: D-2

Mock Midterm 1 - Question 29 | Practice Test 2 Q24

John: half A + half B (positively correlated). Marc: half C + half D (negatively correlated).

Question 29:

Assets A and B have positively correlated returns, while assets C and D have negatively correlated returns. John is an investor who has chosen to invest half of his money on asset A and half on asset B. Marc is an investor who has chosen to invest half of his money on asset C and half on asset D. Therefore, which

Problem 3: D-3

Econ351 Lecture Ch.9 - Variance (formula)

Lecture slide: variance definition and formula.

STATISTICS - VARIANCE

We use the **variance** as a measure of **riskiness**:

$$\text{Var}(x) = p_1(x_1 - E(x))^2 + p_2(x_2 - E(x))^2 + \dots + p_n(x_n - E(x))^2$$

$$\text{Var}(x) = \sum_{i=1}^n (p_i(x_i - E(x))^2) = \sigma^2$$

- ▶ A constant variable has a variance of zero
- ▶ A large variance indicates more uncertainty:

Problem 4: D-4

Econ351 Lecture Ch.9 - Variance (numeric examples)

Compare Investment 1 (low variance) vs Investment 2 (high variance).

STATISTICS - VARIANCE

We use the **variance** as a measure of **riskiness**:

$$\text{Var}(x) = p_1(x_1 - E(x))^2 + p_2(x_2 - E(x))^2 + \dots + p_n(x_n - E(x))^2$$

$$\text{Var}(x) = \sum_{i=1}^n (p_i(x_i - E(x))^2) = \sigma^2$$

- ▶ A constant variable has a variance of zero
- ▶ A large variance indicates more uncertainty:

Investment 1: 50% chance of \$310, or 50% chance of \$290

$$\text{▶ } \text{Var}(\text{Investment1}) = 0.5(310 - 300)^2 + 0.5(290 - 300)^2 = 100$$

Investment 2: 50% chance of \$600, or 50% chance of \$0

$$\text{▶ } \text{Var}(\text{Investment2}) = 0.5(600 - 300)^2 + 0.5(0 - 300)^2 = 90000$$

Problem 5: D-5

Econ351 Lecture Ch.9 - Correlation

Correlation measures whether variables move together.

STATISTICS - CORRELATION

Another important concept is **correlation** between variables:

- ▶ Loosely speaking, variables that move together

Example 1: Daily temperature and sales of ice cream. In days with higher temperatures, more ice cream is sold

Problem 6: D-6

Econ351 Lecture Ch.9 - Diversification

Diversified portfolio (Nvidia + Sony) reduces risk vs single stock.

DIVERSIFICATION

Another way to reduce risk is **diversification**:

- ▶ Allocating resources to a variety of activities whose outcomes are not fully correlated

Value of a \$10,000 Investment After One Year		
	AI Boom Continues (50% chance)	AI Bubble Bursts (50% chance)
Nvidia (high AI exposure)	14,000	7,500
Sony (gaming/hardware)	8,500	11,500

Diversifying to hold both an AI-growth stock and a stock hurt by AI's reduces risk

Similarly, holding a diversified portfolio of stocks reduces risk

- ▶ i.e. buy an index fund, not the stock you 'think' is going to do well

Answer Key

D-1:

a) A & B positively correlated. b) C & D negatively correlated. c) Feng has riskier portfolio.

D-1 detail:

Feng: \$190 if rain, \$150 if drought (EV=\$170, high spread). Nan: always \$170 (diversified).

D-2:

(D) John's investment has higher variance and John is less risk averse than Marc.

D-3/D-4:

$\text{Var}(\text{Inv1})=100$; $\text{Var}(\text{Inv2})=90,000$. Same expected value can have very different risk.

D-5:

Positive correlation: variables move together. Correlation does not imply causation.

D-6:

Hold assets with outcomes not fully correlated to reduce portfolio risk.